

RANA TALHA KHALID

Biomedical Engineer · Wearable Biosensors · AI-Integrated Diagnostics · Biomedical Interfaces
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EDUCATION

B.Sc. Biomedical Engineering · Riphah International University, Lahore, Pakistan 2021 – 2025
CGPA: 3.8 / 4.00 · IELTS: 7.0 Bands · Mukhawat Scholarship (50% fee waiver) · Tameer Scholarship (top 1st–2nd rank in class)

PUBLICATIONS (6 PUBLISHED · 2 UNDER REVIEW · 4× CO-FIRST AUTHOR · 2× FIRST AUTHOR)

Published

- Advances in PVDF for Self-Powered Wearable Physiological Monitoring and Energy Harvesting**
Nano Energy **Co-First Author** IF 16.8 (Q1) 10.1016/j.nanoen.2025.111296
- Reshaping Healthcare by AI-Integrated Wearable Sensors Post-COVID-19**
Chemical Engineering Journal **Co-First Author** IF 13.4 (Q1) 10.1016/j.cej.2025.159478
- Electrospun Nanofibers for Wearable Cardiovascular Health Monitoring**
JSAMD **Co-First Author** IF 6.7 (Q1) 10.1016/j.jsamd.2025.101030
- Next-Generation Wearable ECG Systems: Soft Materials, AI Integration & Personalized Healthcare**
Chemical Engineering Journal **Co-First Author** IF 13.4 (Q1) 10.1016/j.cej.2025.170117
- Hybrid Sensor Integration in Wearable Devices for Cardiovascular Health Monitoring**
JSAMD **4th Author** IF 6.7 (Q1) 10.1016/j.jsamd.2025.100889
- Semi-Automated Framework for Vertebral Measurement in Lumbar Spine MRI Analysis**
Material Proceedings **Co-First Author** 10.3390/materproc2025023005

Under Review

- Bridging Prediction and Interpretation: ResNet-Based ECG Cardiac Age Estimation with Integrated Gradients Explainability**
Sensors **First Author** [Under Review]
- Standardization and Benchmarking of Wearable Biosensors: Toward Regulatory-Grade Devices**
Smart Medicine **First Author** [Under Review]

RESEARCH & PROFESSIONAL EXPERIENCE

- Lab Engineer** · Riphah International University Oct 2025 – Present
- Design and deliver BME lab sessions; mentor capstone projects in instrumentation, circuit design, and signal processing.
 - Established laboratory SOPs and safety protocols; supervise student research and provide structured academic evaluations.
- Undergraduate Research Assistant** · Biosensors Lab, Riphah Int'l University May 2024 – Jun 2025
- Modeled hydrogel-based heart rate sensors in COMSOL Multiphysics; contributed insights directly adopted in the lab prototyping pipeline.
 - Developed Transformer–CNN hybrid for 12-lead ECG arrhythmia classification (92% accuracy); co-authored 4 published Q1 journal papers.
- Biomedical Engineering Trainee** · ENDOKARE, Pakistan Aug – Sep 2024
- Characterized CCD/CMOS sensors in endoscopic imaging systems; prepared regulatory and client-facing technical documentation.
- Biomedical Engineering Trainee** · Doctors Hospital & Medical Centre, Lahore Aug – Sep 2023
- Analyzed cardiac monitors, ventilators, and 1.5T MRI systems; assisted Punjab Healthcare Commission compliance audit.

KEY PROJECTS

- UPEC — Multi-Modal Portable Cardiac Diagnostics Device** *(Final Year Project)*
- Handheld device capturing ECG, PCG, and PPG; hybrid LSTM-CNN classifier (AUC > 97%) for real-time cardiopulmonary disease detection and early arrhythmia alerts — full pipeline from analog front-end conditioning to classification.

Other Selected Projects

- **EEG Motor Imagery BCI** — FBCSP + SVM for gamified stroke rehabilitation **CardioRespiAnalyzer** — MATLAB GUI for cardiovascular & respiratory auscultation analysis
- **PVA Hydrogel HR Sensor** — COMSOL simulation of optical transduction mechanics **Gesture Robotic Arm** — MPU6050 IMU with PID-controlled micro-servos

TECHNICAL SKILLS

AI & Signal Processing TensorFlow · PyTorch · MATLAB · Python (SciPy, MNE) · CNN / LSTM / Transformer / ResNet · ECG, EEG & EMG feature extraction · wavelet denoising · Integrated Gradients · Grad-CAM

Simulation & Hardware COMSOL Multiphysics · LabVIEW · MATLAB Simulink · Arduino · ESP32 · Raspberry Pi · PIC18 · PCB design (EasyEDA) · Multisim · Proteus · SolidWorks · SketchUp

Development & Languages Flutter / Dart · C++ · C# · Python · MATLAB App Builder · English (IELTS 7.0) · Urdu (Native)

LEADERSHIP, AWARDS & COMMUNITY SERVICE

Technical Team Lead & Co-Founder · Riphah Society of Engineering & Sciences (RSES) 2023 – Present

- Co-founded a 200+ member student engineering society; organized 8+ technical workshops reaching 350+ students and 3 industry-academia engagement events connecting students with senior professionals.
- Built a peer-mentorship pipeline supporting 5 student research projects and 2 interuniversity competition wins; received Forward-Thinking Pioneer Award and Mentorship Excellence Award from RSES.

Selected Awards

- 1st Runner-Up, ETECTE'23 (2nd Int'l Conference on Emerging Trends in Electrical, Control & Telecommunication Engineering) — poster presentation on AI-enhanced wearable cardiac sensors for COVID-19 patient monitoring.

Community Service

- Coordinated 34 blood donation drives (500+ units collected for thalassemia patients and surgical emergencies) with Al-Khidmat Foundation — one of the organization's most active student volunteers.
- Active volunteer crisis counselor at Crisis Text Line — completed training in suicide intervention, trauma-informed communication, and mental health first aid; contributes to community mental wellness outcomes.
- Organized Ramadan iftar drives serving 300+ underprivileged families; led hospital clean-up campaigns; 2× interuniversity debate champion.

RELEVANT CERTIFICATIONS

Machine Learning Specialization (Supervised ML, Advanced Algorithms, Unsupervised Learning & RL) — DeepLearning.ai & Stanford University, 2025

Cancer Biology Specialization (Cancer Biology, Metastasis, Prostate Cancer) — Johns Hopkins University, 2025

How to Write and Publish a Scientific Paper — École Polytechnique, 2023 · **Foundations of Healthcare Systems Engineering** — Johns Hopkins University, 2023

Healthcare Data Literacy — University of California, 2023 · **MATLAB Onramp & App Building Onramp** — MathWorks, 2023